SI416: 137 Years of Trend: What the Evidence Really Shows ft. Yoav Git

Podcast: Top Traders Unplugged

AI Executive Summary

Core Thesis

The episode argues that long-term investment success is derived from preparation and trust-based relationships rather than market prediction, reinforcing the viability of trend-following strategies through historical validation. While acknowledging the robust performance of CTA indices over 137 years, hosts emphasize that modern investors must understand the distinct mechanics of ETF vs. direct management structures regarding strategy transparency, and recognize that risk management (volatility scaling) is more critical than raw alpha in diversified portfolios.

Key Arguments & Data

- **Client Relations & Investment Structures:** Direct manager relationships are superior to ETF CTA structures for transparency. Clients need justification for strategy changes during drawdowns to maintain trust, whereas ETF managers can alter underlying strategies (e.g., Simplify CTA) with limited notice.
- **Trend Following Consistency (Peterson et al.):** A 137-year data set covering 67 markets proves trend following delivered positive returns every decade from 1880 to 2016. Performance is driven by volatility scaling and diversification rather than cross-sectional correlation alone.
- **Market Performance Metrics:**
  - **August Returns:** Gresham CT Index up +2.95% (Month), +11.14% (Year-to-Date). Sock Gen Trend Index +2.77% (Month), +10.9% (Year-to-Date).
  - **Benchmark:** MSCI World up +2.48% (Month), +12.13% (Year-to-Date).
  - **Fixed Income:** US Aggregate Bond Index down -20bps (Month), -14bps (Year-to-Date).
  - **Trend Barometer:** Readout at 52 (Strong).
- **Commodities & Oil Dynamics:**

    - **Elasticity:** Commodity demand/supply elasticity is approximately 0.15x (IMF study), meaning a 10% price rise yields ~1.5% production/demand response.
    - **Storage Limits:** WTI storage capacity is approx. 70 million barrels; oil squeeze mechanics are driven by financial player roll obligations rather than just physical storage exhaustion.
    - **Curve Dynamics:** Financial players dominate front/middle contracts (liquidity), while physical players hedge further out based on storage constraints.
- **Risk Management & Governance:** Human risk vetoes should be pre-programmed into simulation logic (database lines) for auditability rather than manual overrides to ensure consistency and prevent post-trade interpretation errors.
- **Correlation vs. Variance:** Trend following performance relies on managing risk units (volatility scaling) to achieve diversification benefits without sacrificing expected return, unlike strategies that rely solely on individual asset correlation.

Actionable Takeaways

- **Prioritize Pre-Brief Communication:** When implementing strategy changes (e.g., volatility targets or new instruments), communicate the rationale and seek client buy-in before execution.
- **Prefer Direct Management Over ETFs:** Invest in direct CTA managers where transparency on strategy modifications is a core requirement to avoid opaque shifts similar to recent ETF management churns.
- **Evaluate on Risk Metrics:** Focus on portfolio volatility, covariance, and diversification benefits rather than individual asset correlation when assessing trend following performance.
- **Understand Asset Class Microstructure:** When allocating to commodities, recognize that price movements often reflect spread market dynamics (liquidity and roll obligations) alongside physical storage mechanics.
- **Implement Systematic Governance:** If human intervention is utilized during trading operations, it must be integrated into the simulation

logic rather than manual overrides to maintain governance standards and auditability.

Full Transcript

Welcome to Top Traders Unplugged. In markets, success doesn't come from predicting what happens next. It comes from being prepared for what you can't predict. In each episode, we go deep with some of the world's most thoughtful minds in investing, economics, and beyond to understand how they think, how they prepare, and how they decide, and the experiences that shaped how they see the world. No noise, no shortcuts, just real conversations to help you think better and invest with confidence. Oh, Cambridge is sunny, but we are getting a little bit of water, a little bit of rain, hopefully. And I'm sitting here sipping my Indonesian avocado coffee that my daughter, who's back from Indonesia, made for me. So life is good. Yeah, certainly sounds like it. I'm not sure I would pick the same coffee, but I haven't tasted it, so you never know. We have to try new things. It's the first time. You have to try new things. That's absolutely true.

Now we've got a few new things on the agenda. Some great topics that we're going to be talking about. We're going to announce the winner of the very important TTU merch competition that was running in August. So I'm sure there'll be some exciting news there. And then we got some papers to review as well. So all is good. However, before we get into any of that, as usual, I'm always interested in finding out What kind of has hit your radar, your desk in the last few weeks that you found interesting? Hi, Neil. It's always nice to be here. Thank you. So actually, I want to talk a little bit about clients, not about current events or anything outside. It's actually, we've had a few interactions with investors in the last month. And it really brought home about the nature of the relationship we have with different investors.

So we have two very distinct meetings. The first one was in Zurich. And we were discussing a location to trend. And the guy said, oh, you shouldn't worry about it because Claude wants us to have more trend in our portfolio. So we've been advocating TPA. It makes sense to be part of a portfolio. And so there's a little bit of total portfolio approach, which is coming along. AI, when you give it the data of allocators, they kind of realize Claude really wants a trend. So I think actually this is really exciting for the industry. I

think that's being dispassionate about what is actually best for the portfolio. It's really interesting. So that was like the first interaction, a very technical.

a very technical discussion. And then the second one was a much more personal. So we had a very long-term supporter of the acre strategy, the commodity strategy. And his retiring is moving off to spend more time with his family. And I thought it was a really, it was a really emotional meeting. And we wish him all the best and him and his family. And it really brought home to me how much, like the nature of the investment, investor relations is really about trust and about the relationship. It's not just, you know, people allocate to you, not just because, you know, you are running a CTA, but also because they trust you. So that, and that was very, I didn't realize how emotional I would get about it. I mean, it's interesting you say you bring that up because obviously that's the side of the business that I've been, you know, working in for more than three decades and i completely agree of course i think it's a it's a very personal business or side of the business very important often of course as you say. People tend to start by screening just for numbers but i think once you get into the conversations do you do diligence and ideally want to start having a business relationship i think that it changes a lot and also find that.

I mean, a lot of it is also about the journey, right? When you've been through a few difficult times with a client and they have felt supported along the way, you've given them the tools and the information they needed to essentially, you know, justify the investment to their investment committee or their colleagues. Once you build up that relationship, that trust as you mentioned, I think, and of course, as long as you don't screw it up completely on the performance side. Yeah, absolutely. I think it really solidifies the relationship and it is often, I think, maybe some part of a business that may not be as appreciated as it probably should be. Of course, I'm biased when I say that because I think a lot of maybe, and this is a little bit unfair, but I think a lot of people who are on the say, the strategy side feels, oh, they're the most important because without the strategy, you know, there's no clients. But on the other hand, I think we could argue whether if there's no clients, there's no strategy. So it's a really... It's a two-way process. And it's really about a conjoined response. I think the

important thing is that you are actually working in partnership with our business development and the investors.

in order to, as you say, they need to justify to themselves and to justify to their IC. And you will go through periods where you are in a drawdown or performance isn't what is expected and they will come to you and you need to be able to actually explain to them what's going on and you're thinking about how... when is the strategy working, when it's not working. So that responsiveness and that sort of direct relationship with clients is something that I haven't done until I came to Gresham. And I really, really appreciate it, I have to say. It's really wonderful. It's an eye-opener about how the business is actually, as you say, running. It's not just strategy. Well, somewhat related to this, and I have to be careful. I don't want to be...

I want to say things that is not fair or reasonable. But I did notice over the weekend that suddenly there was a lot of chatter on X about one of the CTA ETFs out there, actually quite a big one. It's the Simplify ETF called CTA. I mean, it's got a great ticker. And I saw some people talking about it saying, what's going on here? All the positions are being wound down. There's only a few big positions now left in oil. But there was no real news. Then when you looked around a little bit further, some of the portfolio managers had left Simplify recently, Mike Green, one of them, and some of the people who worked on the business development side. But again, no real news, no real announcement.

But then I was someone shared with me, you know, the official announcement that then came out a few days later saying, oh, great news. We're changing the strategy. We found something that's better. I bring it up because partly because we just had the sub-advisor on the podcast, Altis. They're no longer the sub-advisors, as far as I understand. And as I said, they may well have found something that's much better. And, you know, that's great for the investors, of course. However, two things. One, as you kind of suggested, communication with your investors is pretty important. So that's one thing I would just mentioned. But the other thing is in the ETF land, we, I mean, obviously we have people who are great supporters of ETFs on the podcast. I have my biases, of course, as everybody knows.

But one of the things that maybe one should be aware of in investing in these ETFs is that suddenly big changes can happen underneath the hood without the investors really knowing. And even if you get a letter a few days late saying oh we're changing. You know that's a you know how do you really know then what exactly you've invested in and so on and so forth and I do highlight that a little bit even though I may get some angry emails. Because I think that's one difference difference when you go and you invest with an actual underlying manager where they're not suddenly going to completely change the strategy and certainly not without giving notice to the investors. So yeah, I'm just putting it out there not necessarily I don't know if you want to comment on it or not but but I just feel it's important.

And it's an important difference to highlight that when you have these structures that are so convenient for everyone, well, they're also very convenient if you want to change the strategy completely. So, you know, that's my bias comment, of course. So I think it is important that, you know, if you are doing a CTA, you are actually a CTA. But I think your point about communication, I think my favorite, my CLO's favorite word is sequencing. is if you want to implement a change, you speak to your clients first, you make sure that they're on board with it, you explain to them what you are doing, because changes happen, right? You raise volatility targets, you reduce volatility targets, you change, you start trading another strategy. All of these changes, you just basically need to keep them abreast of what you're doing, why you're doing them, and they will be on board. They understand what you're doing. But listen, let's be...

transparent about this or whatever the word is. I don't think if I went to my clients and said, oh, listen, we're going to go and completely change the strategy that they would be on board with that. I just don't think that's going to happen. So I think there are nuances here where you need to be careful in terms of what you mean by changes. Small tweaks, small improvements, sure. Actually, I don't know what's happening. It completely went under my radar. So I'm not good at talking about that. But yes, a CTA is normally a CTA. Well, and anyways, all right, let's leave it at that. All right. And the other thing I noticed, and this was someone mentioned, you know, just a comment, but I did find the soul story, which was in Hedwig. You know, this Bobby Jane had been in the news about him setting up his own fund, et

cetera, et cetera. And then later on, I think he moved into the millennium stable and is managing money for them.

But there was this little notice out in Hedge Week saying that the firm's gross gains since launch equate to roughly 34% of the capital raised, while investors received about 8.4% in net gains over that period. And I thought, hmm, that's kind of interesting. And I know there's a lot of caveats. I'm sure there's a lot of information in detail. But still, those are the kind of headlines that doesn't make. Some of these strategies look great. I think that was part of the rationale for the merger with Millennium, right? So the idea is these days, setting up a multi-strut is an expensive business, both in terms of data and in terms of infrastructure, let alone expenses of what you pay as a pass through to your underlying pods. So there's a lot of friction that's put this way. And I think it makes sense for him to have merged with the Millennium operation? No, it certainly sounds like it. So and hopefully hopefully they will get a higher share of the gross profits as a result. Yeah, no, let's hope for that. Anyways, it is time to announce the winner of the review competition and who gets the TTU merge. Shall I do the drum roll? I did not see any any reviews from you Joav, but the good news is you already have a very prestigious TTU vest already. But before we get into the usual trend update and topics, let me just allow me a few minutes just to talk a little bit about that. So for people who know nothing about what I'm talking about, it is because we had, in my opinion, one of the best conversations about kind of trend and why these strategies make sense between Dave Dredge and Rich Brennan.

towards the end of July, I think the episode came out. It really is a must listen in my completely biased opinion. But we said, you know, this is great. For those who leave a really good review, which of course supports the podcast, let's give out one of these vests, which, you know, right now is really only something that I created for the wonderful people who work with the podcasts and of course all the co-hosts. But of course, once I start looking at these submissions and the reviews, it comes very hard just to pick one because there wasn't really kind ones in between. So I decided to pick a handful. So the good news is that the winners are, and I will email you if you are among them and we'll figure things out, but the winners are Todd, Adam, Anton, Tim and Nikolai. So those are the five winners. And

yeah, first of all, thanks for submitting some really wonderful reviews. I don't think I'm going to read them. I mean, they would be, you know, a little bit too embarrassing for myself, but they are quite nice and positive. So no need for that.

But I also want to say a huge thank you to all of those who left reviews and who did not win this time around. Because it really does mean a lot. And it's also a way to show, as I always end up episodes saying, it's a way for you to show the appreciation for all the wonderful co-hosts that spends a lot of time in preparing for these conversations. So there we are, some great news. And we will have five more people around the world wearing the TTU. bests anyways back to the regular programming as they say so performance was pretty broad based in august which is which is important of course contributions from equities and metals did the heavy lifting but actually there was quite a few sectors that participated which is which is pretty good and i was actually curious from you know from your perspective you obviously trade some alternative fixed income markets, for example, that I have no exposure to and don't really follow. But interest rates is certainly part of the conversation at the moment. And I was just curious, Joav, if you could give us a sense of where are yields around the world, sort of broadly speaking, where are they? Because we always only really talk about the US, but that's just one out of many countries.

Oh, I'm not going to do a full review, I think. But the story is the mark is too, in our portfolio that we are watching. So, Japanese Raita, actually, is the gift that keeps on giving, if you're short, if you're short JGBs. and really... Where are 10 years GDB yields now? Roughly 2.8 if you're in the swap book. The bond book is slightly different. But it's the long data, which is actually at an all-time high. So really quite impressive. Canada got affected by the tariffs. So we are seeing changes there as well, actually on the long side. And then... Malaysia, there's a little bit of a story and China, but really, I think Japan is the main worry at the moment because they financing the...

debt, they've always been easy to finance the debt at 0%. But now that the GDP is sort of debt to GDP ratio has always been high at 250% or thereabouts. Now that yields are sort of the long end is sort of approaching

the 4% level. That's really kind of painful. Yeah. It certainly changes the appetite for foreign bonds like US bonds, I'd imagine. Absolutely. In that sense. And the other thing I noticed for the month of August is that, you know, I think relatively speaking, even though there was a little bit of, how should I say, news that affected the fixed income markets in the U.S. was got business announcements. I mean, generally, I think fixed income exposure in the CTA land was really relatively stable. Maybe equities exposure reduced a little bit because of, you know, a slightly weaker momentum.

if I can call it that. But on the other hand, given the new conflicts coming back again in the Middle East, we have seen some firmer I think exposures in the energy sector. At least that's kind of how I see it at the moment. But as you say, there probably was a few single markets that... Well, actually, in commodities, we're seeing across the board very strong performance. The book is relatively long elsewhere everywhere. And we're seeing things in anything from agricultural, we're seeing in electricity, we're seeing good performance in freight. It's been a very strong month. And the nice thing is about the cross-sectional, the fact that pretty much every sector seems to be sort of push earning it's keeps, so to speak. So yeah, very good. Fantastic. Good stuff. So anyways, as of yesterday, my trend barometer finished at 52. That's a pretty strong reading and that is very much in line with what we see in terms of performance. I'm not going to talk about the first couple of days of September, but the August numbers were as follows, beat up 50 up 2.95% for the month.

Up 11.14 for the year. Sock Gen CT index up 2.49% for the month, up 10.79% for the year. Sock Gen trend up 2.77 for the month and up 10.9% for the year. And even the short term traders index very strong month 2.44% and up 5.7% so far this year. Now in the traditional world, MSCI world up 2.48% that is actually as of last night. and up 12.13% yesterday. Where are we? The US aggregate bond index is down 20 basis points and down 14 basis points for the year. And the S&P 500, so far in September, down 25 basis points and up 11.99% so far this year. Now, we actually had three questions come in for you, Joav.

So let me try and do that. The first one is from Jose and a good supporter of the podcast. So I appreciate that. Now, he did boil the question down to just a few lines. So I'm only going to read out the few lines because there was a longer explanation to it. Anyways, the question is as follows. Does trend following pay too much attention to correlation and not enough to variance and covariance at the instrument level? Given that correlation tells us how stocks spread, while variants might show where stress is becoming most acute. Oh, that's a great question. I've got to tell you, as a mathematician, I love variants. I love correlation. We love them both. We have no favorites children in our book. So I just want to highlight where each one of them is fitted. At the end of the day, correlation is important in terms of how it feeds into the covariance, actually. So a lot of the CTAs, most of us, trading strategies, are univariate, i.e., you look at the price of the asset to predict that.

asset, right? So it's not necessarily cross-sectionally. So correlation features in several places. Let's take the easy one, which is volatility. So we look at a single market, we would basically like to trade in units of risk. We want to have, we abstract the asset and we say, okay, if given our position, what is the number of units of risk and we like to work in units of risk. So we would generally both scale the signal in terms of Z-scoring the trend. So we will take the trend but then divide it by the volatility of the asset to get a sort of a Z-score. But we would also more importantly is we would actually divide our position by volatility, not just volatility by the way, FX or the size of each contract just to make sure that we get a constant unit of risk. So if you're holding a bond in my case, it's the duration of the bond which is important and it's sort of the measure of risk. So that's what volatility is usually used for. Correlation in your portfolio is used in multiple places. I mean, the first place is really about the overall portfolio risk, right? So I've got all the individual assets and I've normalized them so that each of them is a unit risk. What is the overall portfolio volatility. And that's really an important question because most CTAs would manage their book to have a certain level of volatility per day or per annum. So say 15% per annum, maybe 1% per day. And you have to look at the correlation right now between the markets to size that position appropriately. So that's why we kind of care about correlation. But the other aspect of it is that that also gives us the diversification.

So if we think of the portfolio and we have 100 assets and the correlation takes the volatility down from, you know, instead of being 100 units of risk to 10 units of risk, then what we have done is we have reduced by 10x our volatility, but our expectations has changed, has stayed the same. So the square root of the variance, the volatility, that portfolio diversification is what actually gives us the performance of CTAs. So that's why we care about it. It's not so much that we use it as a predictor between two assets. It's really about thinking about where is the risk is of the overall portfolio and how do we manage that to give the investors the return that they want, the volatility that they want.

All right, the next question is from Vladimir. Vladimir writes, when a systematic process permits a human risk veto, what evidence should be preserved so the override can be judged separately from the original signal? For example, would you keep distinct records for a model forecast that authorized a position, the order that reached the broker, and the later outcome? Combining them may hide whether the model failed, the control level layer intervened, or execution changed the result. Separating them may also invite too much post-trade interpretation. I would value your view on where the record keeping becomes a useful governance rather than an unnecessary complexity. That's a good question. So one thing is that we don't really intervene that much.

So it really is important to understand that a lot of like we're not being paid to to be risk managers and to not to be risk-minded to be to make micro call to make judgment. Okay. That's not what we are paid for. We are paid to monitor the risk and make sure that the risk is correct. And most of risk intervention that we do is to avoid left tail and actually cost money to the investor. So we do it very rarely. But actually from your question, it's it's important to understand that even the risk intervention is systemized. And what I mean by that is we would never then tell the traders, oh, don't trade this. What we would do is there is a process to have, there's a leverage that we already set within the simulation that will tell the system, don't increase this position, right? So the system can load from a database an intervention and then it will trade the way it trades. It's incorporated into the code. So it's never a case that we manually override something. We program into the system, into the simulation, and interventions, the interventions that we are

likely to do, reduce the risk of the portfolio, reduce the risk of a certain asset, stop trading, stop increasing. All of these, you pre-program it. You don't do anything which you don't. So you're always able to play back because the process is, I think there is the Homoose crisis in Iran, right?

So I think maybe we shouldn't increase the positions in oil, okay? It will go through the CRO, we will get an approval, and then we would basically add a new line into the database, which is this intervention. That would be loaded by the simulation, and the simulation would know not to increase positions in oil. That's kind of how it works. It is really important, as you say, in terms of governance and later auditing what's happening. We definitely also monitor, by the way, the execution. So sometimes if you get a partial fill, we need to know about it. Any drift between the simulation and the actual position, we have to monitor. And we need to understand how much risk that is amount and what happened to that and why is that happening. So all of these are really important. But I think it's important to understand that the simulation is a tool that does many things and it's important to separate what they do. And for each question that you ask, it will be a different answer. So like we ask, we look at the simulation as one way to look at the simulation is to generate what should be my position today. Okay, that's really important part of the simulation. But we also use this as a statistical question, which is, you know, is this code change actually makes sense? And there you have to answer like separate question. There is a like the difference between is there a gross alpha in this signal?

Okay, there's a difference between is correlation calculation correct. There's a difference between is what is the trading cost? Can can we actually bear the costs of a particular fast say fast trend predictor with this market? All of these questions are very separate and the simulation is not just one tool It is designed so that you can tweak it and you on and you set up the parameters of the simulation to answer the specific question that you want to answer and that's something that people don't really appreciate. Yeah. Yeah. And of course, all firms do it differently. I think for a long time, I think generally, there was a school of thought among established CTAs that it was very rare that you would intervene, however. And I mean, discretionarily change things. I know for sure we certainly don't do it over at Don. But I do know that some managers came out after, say, COVID.

and was very open about it said that, yeah, we did intervene and we did reduce the risk and so on and so forth. But I think as long as people are transparent about it, and as you say, probably a good idea to pay attention to it and log it and all that stuff, it makes sense. All right, final question from Bandazian. You talked about different market participants leaving different footprints along the curve. Where today can you most clearly see hedges, CTAs and discretionary macro players fighting each other? Oh, I love this question. And the reason is because the question, this is follow up for the question last time that I was on TTU. And I think it was Marcus who asked exactly this question about the different points on the curve. And I think we also have a paper.

exactly addressing this question. But I'll just answer it generally. It really depends on the commodity market and the liquidity in the market. In general, where you will find most of the macros and the CTAs and what I would call the financial players, you will find them in the front contract. This is the one which is most liquid. You will see physical players playing in the sports market, so actually cash. cotton or cash wheat or cash whatever. And then there will be producers and consumers that will be hedging out or further out the opposition further out on the cliff. So if I'm an oil producer, I'm likely to sort of forward sell my oil sort of.

sort of towards three years, five year points. So that's kind of the broad brush stroke picture of who's playing what. Okay. Very good. All right, now we are back into the topics that we usually bring up. And there are a couple of main papers I think we want to talk about. And I, as I mentioned to you, to be completely transparent about it. It's month in, we're recording very early in September, so it's been pretty busy on my side. So I'm going to let you lead the way through both papers here. But the first one is, of course, a classic. If you have come from the trend following world. And let me just point out, there was another Dane.

involved in the paper, so that makes it even more special. Maybe we should ask him to join the podcast one day if he's interested. Anyways, we are of course talking about a century of evidence on trend following investing. Very famous paper. I know Lester Peterson was involved. Who were the other authors to remember? Oh, I can't remember. Some people from AQR.

Yes. Which is really embarrassing because I really should know who the people are. But I don't remember. I think Hearst was another one. There's Brian Hearst and there's Yahuwah. Yes. I think those are the three. Yes. All right. Anyways. Over to you, my friend. Oh, yes. So this is a classic paper and it came out and it's really like one of those papers that come out and you think, damn, I really wish I wrote it myself. Because they've done a great amount of work. They've basically taken, created a 137-year data set. They've looked at monthly return across like...

67 global markets, major asset classes, commodities, equities, bonds, and FX later on. And they've like literally hand collected from Seabot, they've hand collected a lot of historical commodity data. And with their explicit aim to check whether trend following works, right? So in I think 2012, Lasse-Petersen came up with his sort of TS mom paper, the famous one. And this one is like basically, let's go back and see, is it a recent phenomenon or is it something which goes back throughout history. And it's not going to be a surprise to you that trend following is one of the most consistent across all asset classes and across all decade. They've gone back to 1880 and guess what? Every decade was a positive return from 1880 to 2016.

But they've also looked at other things that are important for trend followers. They've looked at basically the smile, the Christ Saffa. They've looked at stock bond diversification. And what I found is that pretty much in any environment, any macro regime, whether it is in high inflation, low inflation, or whether it's a boom or a bust, whether they're even through major wars, trend followings just seems to work. Obviously, in any given year, any given asset, there will be losses. But overall, it's an amazing result. And the reason I'm putting all this paper, because it's been essentially 10 years since they've done it, so we've got another decade in the mix, and I thought, oh, let's have a look. By the way, there have been papers afterwards that have criticized it, which we can address it as well.

But I think one of the more important aspect is that the sheer body of evidence supporting Trent, right? So I think my colleague Tom Babbage, he wrote a blog recently about, he's an astronomer and he's looking at the way that you examine at distant galaxies and you have an image which is very

noisy and it's very difficult to detect the actual star. So, but what you do is you juxtap, you take many, many, many photos from different places and over different times. And over time, the noise disappears and cancels out and the phenomenon appears. And that's exactly how trend works. We've got a, any individual asset, any individual year, the sharp is not great. In fact, the noise level, the signal to noise ratio is very similar for like a distance star and trend following as a signal. But SU, SU, diversify in terms of the asset classes and in terms of the years, a clear picture emerges. And I think that's a really beautiful analogy that Tom uses. And it held up, the good news, it held up another positive performance this decade. So we've seen this another 10 years of past and trend has held up. 2017 was an outstanding year, but there have been good years also during the COVID time.

there have been good years during the Ukraine crisis, during inflation crisis, pretty much in all regimes, trend is actually done decently. And it's really delivered on what it was supposed to do. But I think we can go deeper, not just in terms of... Yeah, no, I'd love to go deeper. Well, let me try and interject just a few things and thoughts that comes, brings to mind. Now it's good to know that since 2016 when the paper data series ended that the next 10 years have been fine but we have also been you know quote unquote criticized as an industry for. Lower returns in the last 20 25 years compared to the 70s the 80s we've been obviously the 2010s people kind of picked that and said all that.

you know, it doesn't work anymore and so on and so forth. Now, I know that, you know, that we interchange trend following with CTA. And of course, that's not exactly the same. The paper obviously talks about trend following. So that is different from quote unquote, the CTA indices that some people follow. And, um, and I don't know if this is true or not, but even within quote unquote, trend following programs, maybe there's Little bit of non trend that has snog in during the periods where trend didn't work so well. But you know now that we sit in 2026 10 years later from the paper. Some of that criticism about returns being lower and so on and so forth is that in your opinion without it being too much of a leading question is that because we're and I brought this up.

on in my last conversation with Tom and Andrew that maybe the indices we're tracking are not really a great indication of the true performance because all the bad strategies that eventually disappear from the index are simply dragging the returns down, which is why I would say, you know, investors should really. try and find managers that are able to outperform the index. I mean, we should be able to outperform the index in my opinion over time. But anyways, I mean, I'm rambling a little bit here, but are there any thoughts now that we sit 10 years later towards some of that, quote unquote, criticism that we've received? So I have to say, for what you're receiving, you would normally pay for.

You think about trend that's actually buying protection. It is a defensive strategy. There are very few strategies out there which have both a defensive nature. They have positive convexity. In fact, if I plot all the HFR hedge fund indices, right? So normally you would kind of pay for that. You would expect to pay. And in CTA land, we have very strong evidence that you've made money over the years. So I think it's in a weird way. I think CTA is pretty much the only one of the, not the only, but one of the very few phenomenons that has been super consistent over many years. It's very understandable what's happening and actually provides you positive convexity versus sort of just Tapping into sort of liquidity premium. The funny thing is here you have search interrupt. The funny thing is that I agree with you. Normally you should pay for this, right? But in our industry, it's almost become like, oh, you don't have to pay for trend. It's so easy to do. You don't have to pay for it. That's kind of the funny thing because it's actually... No, absolutely. I think when I joined the CTA industry, the trend industry, then It was considered like an alpha. And of course, as the secret source has been, people know it and we are actually indicating investors about it. Then, yeah, there are more entrance to the field. But I think what really begs the question is that the AUM of the industry hasn't really gone up. So it might be that it's gone over to the pods inside the multistride and so forth.

But QIS products, ETS, absolutely. And it shows you that it's about supply and demand. The reason why people are not willing to pay for it is not because they don't think it doesn't work, it's because they think, oh, I can get it cheaper. Which is fair, which is a fair point. So it's not just how good

your product is, it's also... You know, it's also the fact that there is a lot of competition for it, right? So that's why, you know, price of electronic consumer electronics has been dropping. It's not because consumer electronic has been getting worse. It's just that, you know, we're just becoming very good at, you know, a competition. You can buy, you can buy a Samsung, you can buy an Apple, you can buy the plenty of, there are plenty of providers of, you know, a TV, a consumer TV. No, there is, there is, there is genuine phenomenon that we've seen in the last 10 years, which actually the paper sort of hints at. So if you look at historically, decay of performance, there has been a decay of performance. The markets were a lot more liquid. The markets were a little a lot less efficient in the good old days. So one would expect to see some deterioration over the years. By the way, I'm pretty sure that if you were to run, if you were able to run relative value equity, some sort of long short equity fund historically, which is very difficult to do, you would still see the same phenomenon exactly with any hedge fund strategy. But actually, I'm sorry to interrupt again, but I just want to challenge that a little bit because I think that's what a lot of people would say. Oh yeah, just look at the returns, they're much lower today. Yeah, but so is the volatility, right? I mean, I wonder if you valid just returns for the drop in volatility. I mean, when I started in this industry, I mean, back in the 80s, we had the CTA, even the Barclay CTA index had a vol of more than 20%. Right? Now it's down less than five. So... Well, personally, I mean, I measure things in sharp because we can provide pretty much any volatility that the client would want. Right? That's the whole point of what we do in terms of leverage. But...

What I would say is that 10 years is actually a very short time in terms of making a statistical decision in a weird way. So you will find if you're a new quant, if you're a young quant just going into the industry, they don't believe in trend because the attention span is like two years. But the point being is that actually in order to see data clearly, in order to see a certain phenomenon clearly, You first of all need to understand the fundamentals, secondly, why this is happening, but also the amount of data involved is substantial, which I think is why this paper came about. So I would say variation in performance in the last 10 years versus the historical, you can put it over the historical performance, a little bit lower. Yeah, that's fine. Still positive, definitely.

you know, faster, faster seems to be degraded in the last 10 years more than slower. Yeah. But you know, what can I say? I think the one, the one, the one, the one thing I actually do want to go back to is the correlation story and the diversification. Okay. Which is the question from Jose. Yeah. One of the findings of the paper is that the system performs well, trampoline perform well when the assets are uncorrelated. Yeah. And that's exactly what we're talking about. The more diversified your portfolio is, you can load volatility while keeping the expectations, and then you basically level things up to get a higher return. And over 130 years, the more uncorrelated your assets, the better your performance is. And that's something that people don't normally think about in CVAs. People think about Assets crisis alpha and in fact, you know, we do Do things you do well when there is a crisis alpha and I think it's important to understand that the correlation of the trend PNL's and The assets are not exactly the same thing, right? So we do like it when all just like we had in August when all Sectors are firing for different reasons potentially, right? That's that sort of correlation. We actually like in terms of the performance PNL market correlation we don't like, we like low correlation because then we actually able to leverage up the portfolio and get a higher return for the investor. You know, you brought up something that I don't think it was intentional as such, but actually it's a really, really important point. And that is your entry point to this industry makes a big difference of whether you're a believer in trend or not. I actually think that because I believe that people who Probably experienced the say the first eight year well anything before 2008 probably say yeah, we're believer we've seen it work many times if you came in kind of in 2009 and you went through quote-unquote the quiet period with not too many years being kind of super years I think you are much and on and well It's just a fact that that a lot of a lot of the decision makers nowadays are people who've only been around for 10 or 15 years. And so they are likely I think in my opinion. To believe less in in this strategy vice versa they may be maybe strong believers in in in a in a long equity type strategy because equities have been on fire so so the the.

And I think also, even in my own case, sort of getting into the financial world in the mid 80s, you know, the whole 2000, sorry, 1987 crash certainly probably made me much more cautious overall, even though people might think, well, that's crazy. He's involved in trend following. How can he be

cautious? Well, actually, I think trend following is a pretty a very well-risk managed strategy, we are risk managers first and foremost. So I feel much safer in a long, short, fully diversified strategy than I do in a long, only strategy. Absolutely. I think if you look at the variability in volatility in equity markets, it's enormous, right? You can go through very quiet periods and then volatility spikes up. We get paid for that. And I think one of the, it's quite funny because one of the biggest criticism, there have been papers criticizing this work. And there's a famous paper, which basically criticizes to unfollowing it says, it's all about volatility scaling. Okay. The risk management.

is what the performance is coming from and to me is like precisely what we do the reason why we're able to essentially abstract each of the markets and actually put them together in a coherent fashion is because we manage the risk and we because we've volatility scale to get you know proper diversification of the portfolio and it feels odd to people criticize oh You know, it's like saying, you can't make, the cake was only good because you put eggs in it. Of course, but you know, that's part of the recipe, right? You can't criticize that, right? It's completely integral to what we do. So, you know, that's harked back to the first question that we asked. You know, other people criticize trend following because of costs. You know, they look at the simulation Pettus and Dad did and they said, oh, you know, the costs were higher than that day.

That may be the case, but again, simulation is about answering specific questions. And the question is, is the gross alpha there? We can discuss later whether it's relevant today in terms of being able to extract in today's costs level, whether I'll be able to extract it. So it's a different question to ask. I think this paper has been a wonderful paper. And I think investors, you're right. Investors who actually care for long-term performance, and they care about allocation for the long-term growth of capital, then CTA and maybe TPA and Claude will help you to do it, will actually allocate quite significantly to turn following. I'm excited about Claude, I have to say, because I think Claude has some friends in the open AI world and any other AI, because as you see, When you take an unbiased look at these types of things and you want to build the quote unquote best portfolio, you definitely need, as probably everyone realizes on this podcast, that you definitely need

a meaningful allocation to this wonderful strategy. Okay, anything else you want to mention here before we move on? By the way, I think our friends over at CFM, they did a Yes, 200 years. Maybe that's the one you need to bring next time. It's absolutely right. There was a period when people were going stretched back as long as possible. I think did Katie and Alex in their book do 800 years or something like that? I can't remember. Maybe they did. I think what is really interesting is that I did it for...

other asset classes like option trading and stuff like that. Yeah, I mean, you don't necessarily publish it, but it works. And the reason it works is because we understand why it works. There are two effects that drive trend. The first one is behavioral, right, in terms of people by winners. And we understand that. It's exactly why people don't believe in trend because it has had a few weeks, a few... sort of weak years. That's fine. They're literally trend following and demonstrating why trend following works. The other reason, of course, is supply and demand, which is the mismatch between supply and demand is actually what drives prices. And the fact that price adjustment takes time, that sort of auto correlation in the price really drives what trend following works. It just makes sense. Let's move to something different.

But also about supply and demand, as far as I can tell. Again, I'm going to leave you to lead our way through this one. OK, so this one is about Lea Boucher. I hope I pronounced the name correctly. It's a stylized model of the oil squeeze. So for those of you who don't remember, COVID happened a few years back. And in April 2020, crude was essentially long squeezed the holders of the long of the holders of the WTI contract had to Pay for people to take the contract of their hand and oil printer that a negative a negative print Okay, so the front contract Traded negatively which was a shock a shock to the system and Ilya Bushev is like a very illustrious, both a quant and a manager of energy portfolios. He's published a really good book called Virtual Barrels about the oil industry. And this paper is a short paper that he wrote, which is amazing. I mean, if you're an oil trader, or anything like that, then you really want to read this oil quant. You really want to read that to read that paper. And I was the name of the paper. So the name the name of the paper is sort of a stylized model of the oil squeeze. Right. And what is this paper about? So it has like three distinct parts and is

getting to the covid crisis. So the first one is reviewing the the theory of storage.

That's the first section and it doesn't like it. So normally we have supply and demand in economic theory and what you have is inventories which make up for imbalances and they can do that until you run out of storage. At that point, you switch to a regime where supply and demand really have to do all the lifting themselves. And then instead of price stops being sort of relatively small movement to be very volatile. And that kind of predicts very strong positive skew because the point where you run out of storage is the point where the commodity is in high demand. And basically the price goes spikes up because people run out of things. And we can see that in oil, for example, recently when, you know, In Asia, people would pay 200 to just have a barrel of oil because demand and supply in the oil industry is very inelastic. Can I just... Sorry, I want to make sure I heard this correctly. You say the time when things run out of story is when it's in high demand, but actually with COVID it was the opposite, right? Exactly.

In traditional storage theory, you are in once, you know, when you run out of, run out of, run out of any, any inventories, stocks going down to look at what happened in oil is potentially, it's not quite a story, but potentially on the other side, which is what happens when there's too much, there's too much supply and you cannot store it anymore. That's actually one of the things he points out. That's actually not what happened, but it's the common story people were saying. So that's interesting. But what he's pointing out is that in this model, there is a right skew for the commodity. And we see that in agriculture. So for example, we sell coffee, spiking up cocoa, spiking up when there is actually a shortage of the commodity.

But he says, well, actually, if you look at oil, it's that way round. Oil is essentially the volatility of oil is constant. Whether it's 20%, if it's trading at 20, the volatility will be around 20 annually. And if it's trading at 100, the volatility will be around 20 annually. It's in a percentage term, it's actually inverted. So it's got a strong negative skew in many ways. And he's saying, this doesn't make sense to us. What we really want to do is to skip demand. function altogether, and we're going to model spreads as a direct function of the capacity utilization, exactly this point about storage. We look at how

much of the storage is available, okay, and on the one side when it goes to zero, when capacity is utilized, we have the spike going off to infinity, but on the other side when capacity is fully utilized, then we should go to minus infinity.

right? Price should go to minus infinity. And then we kind of use a bit of a hokey pokey magic sort of expectation forward expectation by saying actually we can think of the capacity utilization as a random variable which mean reverts okay which it does which is correctly and we basically think of integrating that plus infinity at zero and minus infinity at one to the current levels. And you get a nice shape which is you know, as a nice spike near zero to the upside and the negative spike where capacity near one. Okay, so it's a really nice model, so heuristic model and he calibrates the model to actually fit the oil spreads that we see in the market. And then we can think about, he talks about COVID because COVID what's actually happening there, we didn't actually run out of capacity.

And he's talking about a specific incident about the Chinese retail investors and about the way they've been unable to essentially hold the role during the role. So it gets complicated if you want to read about tasks, trade at settlement prices. But I think it's one of those papers where you know that this person is both a quant, so he's able to come up with a model, and he really understands the dynamics of the oil market. And it's a beauty to read. And I really love this paper. And I'm going to qualify it because I've got a disagreement with it as well. OK, let's hear it. So let's do the good things. So you really understand the market, you understand the participants, the physical players. He really explains the carry trade. One of the things that I really have a bug about with lots of quants in the industry is, and by the way, we found last time, paper which he wrote about Carrie which is yeah yeah yeah no no no it's it I think Carrie in commodities and Carrie in fixed income are very different beasts Okay, because money is fungible, you can move it easily from side to side. In commodities, the way that Kari interacts with the inventories and with the capacity, with the storage mechanism, really important. Like if there is no oil, you cannot borrow it, right? It's a fixed commodity. And really, most quants really simply fail to recognize how Kari trade is in tights to inventories. And in fact, you know, if you plot capacity utilization against against the spread against the contango versus

backwardation. It's a very clear line. It's a completely sort of spread is almost is a very, very good function of inventories. So and it really understands the dynamics of inventories about the way the traders actually look at inventories now, but also look at what it's likely to be because supply and demand in the oil market is very inelastic.

Okay, so if we have an imbalance this week, we are likely to have an imbalance next week. So the fact that this is, you recognize that, that's really beautiful paper. And I think we'll come back to the COVID explanation is 100% correct. I just want to run a back of the envelope with everybody, which I think will be a good explanation. So there's a great IMF paper, by the way, a 2024 paper talks about price elasticity in commodities. Again, it's called the power of prices and how fast do commodity markets adjust to shocks. We talked about GIV, about calibration of supply and demand when we talked about equity markets actually a few months back. And they've done this exercise and they've done it for all for commodities on an annual basis. They've looked at anything from bananas to oil markets to copper. And what comes up is that oil is indeed quite inelastic. It has about a multiplication factor.

of 0.15. And what it means is that if we see a 10% rise in price, we will see production increasing about 1.5%. And we will see demand, the number for demand actually is quite symmetric. We will see demand following by about 1.5% for a 10%. And so now let's look at the oil industry as a whole. We start the Iran crisis. with about somewhere between, depending on how you quote it, somewhere between 86 and 100 million of oil being produced every day. And we started the crisis with WTI roughly at 60, a little bit above it. And what we know is that there are 20 million barrels of oil going every daily through home moves. And if you lose 10 million barrels, which actually is what we got, then what we see is In order to balance that, we would need to see a 33% rise, which will give us a 5% decrease in demand and a 5% increase in supply. And that will make up roughly for the 10 million barrels of oil we lost. So the new stable equilibrium would be at 80. If you lose the whole 20 million barrels per day, the price would have to go to 100.

precisely to get the supply and demand sort of balanced. And I think what's really interesting is that this study, which was done by a bunch of academics from the IMF, okay, based on annual data and you might think it's not accurate, actually came out pretty spot on, right? Because we've seen WTI trading in the 80 to 100 dollars range. And I think it really shows you how understanding price elasticity is really important for our purposes. Can I just, before I forget, I mean, yes, but I mean, that it's keeping oil. And I guess that is a little bit of surprise that we haven't seen more than more than 100 each, you know, in oil. I mean, it did require a lot of reserves to being released. Absolutely. And, you know, there are still I remember our conversation a few weeks ago with Adam Rosenzweig. I mean, it's like a billion dollars that's gone out of the market. And in his opinion, you know, it takes a few months before it shows up as as missing. I know the Chinese stopped doing exporting a product. So that obviously helped a lot. But still.

So what you're describing is exactly the events that bring... So that price change is precisely... So the Chinese stopped consuming oil precisely because it was becoming more expensive to them. So the story, the economic theory... is actually being played out in the market. But reserves is actually a great point, right? Because what the US has done is essentially bringing more oil to the market, increasing the supply. They reduced inventories by about 120 million pounds, which is not insubstantial, but also elsewhere. So I think that the discussion about inventories and about how these are managing that process, because what we are seeing is that the front of the curve has gone up, but if you go five years out, you remember we discussing five years out players, right? The price is still a 60. Okay? Like it hasn't, it hasn't budged. Okay? So, you know, I think it was pretty much spot on from what it was then. What it, if you look at the shape of the curve inventories are allowing us to essentially to delay that shock and that plays out in the shape of the curve, which used to be, you know, the front spread when the crisis started was something like 40 cents. It's now sitting at sort of three dollars, three and a half dollars. So it's exactly that. That's the way that the crisis has played is through a much, much, much steeper curve, but correlation of the curve in the front.

So we are seeing fronts sitting at sort of 90 and the back sitting at the back of the curve five years out still hasn't budged till at around 60. So

inventories are a very strong part of that story. But I kind of don't... There's one issue which I have with this paper, which is there's a little bit of a magical switch in the paper because it starts talking about... It starts talking about the oil in elasticity, which I completely agree with. We just looked at it. It's a very small number, 0.15. But then the second part of the model, where we fit it, it fits the spread. It looks at capacity utilization and is fitting the spread. And the spread is a different beast. And in fact, the spread is actually a lot more elastic.

because then you have different players in the market. And I think it's really important to understand supply and demand, not an oil, but an oil spreads. And that's actually where the action is because the players are there, a lot more elastic. But they still, they still obey economic theory because what you have is you have the tank storage, tank farms. In WTI, there's, I think, about 70 million barrels of capacity there. They are a lot more dynamic to the shape of the curve. For them, the price is exactly the role yield, the difference between the front curve, the front price, and the front contract and the next contract. And there are the suppliers of inventories. And then on the other side, you have the misbalance between producers and consumers like refineries on the demand side.

And the dynamic between them, we can actually model very much like economic theory with supply and demand. And it makes perfect sense because they are very sensitive. The physical players are very sensitive to the level of inventories because I can only offer storage if I have storage left in my tank. So as the amount of storage I have in my tank sort of goes down, so say in this crisis, I become more price in elastic I will require a higher spread okay to release oil from my tanks okay and similarly on the other side of the of the equation when we reach capacity like in the covid sample I will require much much much stronger contango in order to store it in my tank Okay, so the supply and demand works perfectly if you think about it, not as an oil market, but as a spread market. But I think the spread market is really also very exciting in its own right. Because there's a bunch of people that trade spreads, they don't even know they're trading spreads.

Okay, and I'm talking about the CTA markets, the CTA people. So if I ask you, do Dan trade spreads and if the clients would come to you, do you

trade spreads? You will say, no, of course not. We just trade out rights, whatever. I mean, there are CTAs that specialize in, there's Moritz and Moritz who are very much into spread. But broadly, we just trade the outright oil. If I ask you, if I'm a long only investor in oil funds in the USO, Are you a trader in spreads? You would say, of course not. This is just long only oil. But actually you do trade oil. You do trade the spread. And you trade the spreads, but only when you roll your position. So what you have is you have physical players who play the spread market and they do it during the month and everything is fine. And then we're getting to the point where we need to roll the contract. And suddenly you have an influx of financial players.

And they are truly price insensitive. They don't even realize how price insensitive they are, because they just don't want to get delivered the oil, right? You know, the CTA will never hold their position, a long-only position. And at that point, what you have is you have a bunch of financial players coming in with excess, if you think of it as a demand shock into the spread market. And the physical players have to make up for it. And normally in normal operation, this is great. But when they are not, like in COVID, you have the financial players who are 100% must roll the position. And that's exactly the incident that we saw in COVID. So it wasn't the case that the physical traders were running out of space to store. But it was the case that the amount of demand shock that we saw.

the role wasn't matched by the physical player and suddenly you saw the big spike in the contango and the price went negative. So I think you can really explain this as a supply and demand but you just have to be aware of like this is not the oil market per se, this is the spread market. Good stuff. I think we have to wrap it up because we are past the hour by some margin. Can I just say something about trend following? Just for this, because we talk about the physical markets, and if you're a CTA, you're thinking, OK, this is a really nice story, but it's not relevant. But I think it is important for people to realize that, which is the first one. The first lesson is that for CTAs and for any investors, The role is an issue in most liquid commodities, but if you're trading alternative commodities or slightly less liquid market, it's not just the daily volume, it's also the liquidity during the role, because you don't want to be short-squeezed or long-squeeze in the position, so role is

really important. And the second question is really when you understand the dynamics of the market, trend following in commodities is not necessarily asymmetric process, right? There is a some assets will have negative skew, some assets will have positive skew. And you really want to think about how you want to handle it. And good luck with that. And I think more it's a noise probably know all the ways to know about about spread about trend following in spreads. But I think it's really interesting in terms of understanding the dynamics of who are playing there in terms of creating understanding how to harvest trend in spreads.

Sure, sure. I think the most important thing actually might take away is that when people think about our trend following is easy because all you need to do is develop an algo. No, I mean, I think we have to appreciate the fact that the experience of the people who implement the trades, especially in the commodities is hugely important. And that is, as we talked about very early in our conversation today, there are so many parts of a CTA business. that needs to work together in order to make the whole, you know, the best it can be. And that also goes in this case. So great stuff, very interesting. I've certainly learned a few things here, you also very much appreciate that. And by the way, even though the competition for leaving a great review is over for now, I will encourage you to do it anyways, because It's a way to show your appreciation for your and all the other co-hosts for the work they do and all the preparation. So please go to your favorite podcast platform and leave a rating and review. Now, I think that's pretty much it. You can of course always go and check out the website for all the updates we do every week, blog posts and the trend barometer to follow along in addition to tuning in each week.

Next week I will be joined by Rob who comes back from his very extended holiday that he always takes in the summer. No doubt he'll be full of energy and you'll be a very insightful conversation I'm sure. If you have any questions for Rob please send them to info at toptradersonblock.com and I'll do my very best to get it in front of him. That's it for now from you over me. Thanks ever so much for listening We look forward to being back with you next week and in the meantime as always take care of yourself and take care of each other Thanks for listening to Top Traders Unplugged. If you feel you learned something of value from today's episode, the best way to

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